

Anshula Gandhi

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Education

- 2023–2026 **University of Cambridge**, *PhD in Math*, Supervised by Timothy Gowers, Thesis: (exp.) Mathematical Proof Generalization and its Applications.
- 2015–2019 **Massachusetts Institute of Technology**, *B.Sc. in Math and History*, 4.8/5 GPA.

Research

- Spring – **Fulbright at Czech Technical University**, *Fulbright Research Fellowship Grantee*.
- Summer 2021 Worked with mathematicians to develop tools for automated deductive reasoning when applying the polynomial method. Focused on using automated-theorem-proving tools in combination with human insight and ingenuity to prove mathematical lemmas.
- Spring 2020 – **MIT Center for Brains, Minds, and Machines**, *Senior Research Assistant*.
- Fall 2020 Developed neurally-guided program synthesis techniques to solve a variety of symbolic reasoning problems through applying the neuroscience behind how humans do math.
- Fall 2019 **Universidad Nacional Autónoma de México**, *Visiting Researcher*.
Developed reinforcement learning algorithms to prove theorems in lattice theory, and developed a Coq tactic to apply duality to prove theorems in a single step.
- Spring 2019 **MIT Center for Brains, Minds, and Machines**, *Undergraduate Researcher*.
Designed reinforcement learning environment to prove math theorems in group theory. Combined machine learning and automated theorem proving to design an artificially intelligent agent.
- Spring 2016, **MIT Distributed Robotics Lab**, *Undergraduate Researcher*.
- Fall 2017–Fall 2018 Developed algorithms to determine safe advisory speed for parallel autonomous vehicles, given locations and speeds of other cars. Explored mathematical path-planning algorithms such as harmonic potential fields and visibility graphs. Constructed a closed-form cost function to determine the riskiness of moving in a certain direction based on surrounding density and velocity data.
- Summer 2016 **MITRE Nanosystems Group**, *Research Assistant*.
Designed circuits and algorithms for novel non-invasive medical diagnostic tool to reduce size, weight, and required power. Conducted chemical laboratory tests to correlate quantum dot fluorescence in the device's sensors to the presence of analytes.
- Summer 2015 **MIT Exoplanet Theory Lab**, *Researcher*.
Devised and coded algorithms to classify chemical spectra into strong and weak biosignature gases in order to pave the way for detecting life on other planets.
- Summer 2014 **NASA Goddard Space Flight Center**, *Research Assistant*.
- Spring 2015 Developed microlensing-based exoplanet detection algorithms in C++ to be deployed on NASA's WFIRST telescope.

Publications / Accepted Talks

- 2025 **Joint Mathematics Meeting 2025: Special Session on Mathematics Informed by Computing** *Refining Conjectures via Automatically Generalized Proofs* by Anshula Gandhi, William Timothy Gowers, and Anand Rao Tadipatri. We showcase a methodology used by research mathematicians to generate intermediate conjectures, and present an algorithm that can help aid humans in this process. We provide concrete examples of how mathematicians often generate conjectures, generalize failed conjectures, and use the generalization to formulate a refined conjecture, and we explain the implications for math education.

- 2025 **Interactive Theorem Proving 2025** *Automatically Generalizing Proofs and Statements* by Anshula Gandhi, Anand Rao Tadipatri, Timothy Gowers. We present an algorithm implemented with the Lean theorem prover that takes in a theorem and its proof, as well as a term to generalize, and generates a more general theorem that requires only the necessary hypotheses on the generalized term.
- 2020 **NeurIPS 2020 Workshop: Beyond Backpropagation** *Supervised Learning with Brain Assemblies* by Akshay Rangamani and Anshula Gandhi. We propose a new supervised learning model based on a network of neural assemblies that learns through Hebbian plasticity instead of backpropagation.
- 2020 **NeurIPS 2020 Workshop: Learning Meets Combinatorial Algorithms** *Dreaming with ARC* by Andrzej Banburski, Anshula Gandhi, Simon Alford, Sylee Dandekar, Sang Chin, Tomaso Poggio. We propose an approach to solving mathematical abstract reasoning problems using program synthesis algorithm using a DSL made of human "Core Knowledge" priors.
- 2019 **ICRA 2019**. *Dynamic Risk Density for Autonomous Navigation in Cluttered Environments without Object Detection* by Alyssa Pierson, Cristian-Ioan Vasile, Anshula Gandhi, Wilko Schwarting, Sertac Karaman, and Daniela Rus. We introduce a closed-form vector equation that allows a car to navigate its environment without explicit object detection and movement, and experimental verification of validity.

Teaching

- Winter 2025 **Self-Employed**, *Mathematics Tutor*.
Worked one-on-one tutoring a blind mathematician, using a combination of braille, tactile diagrams, audio graphs, and audio description to make A-Level Maths more accessible.
- Fall 2023 **University of Cambridge**, *Undergraduate Supervisor*.
Worked on practice problems with pairs of students taking the Groups course at Cambridge.
- Winter 2023 **University of Cambridge**, *Undergraduate Supervisor*.
Worked on practice problems with pairs of students taking the Real Analysis course at Cambridge.
- Spring 2021 **Boston Partners in Education**, *Math and Reading Mentor*.
Worked within a second grade class at Boston Public Schools to provide extra one-on-one attention and tutoring for students.
- Spring 2019 **MIT Combinatorics Seminar**, *Student Presenter*.
Presented a series of three talks on Combinatorial Nullstellensatz, a discrete mathematical proof technique, as a student in MIT's *Seminar on Combinatorics* course.
- Summer 2017 **MIT Educational Studies Program**, *History of Mathematics Course Teacher*.
Taught summer course to middle schoolers on the history of mathematics, including the history of infinity, lotteries, and computing.
- Winter 2017 **MIT Development Lab**, *Electronics Workshop Leader*.
Co-led month-long workshop on circuit building and microcomputer programming, as a student in MIT's Development Lab. Worked with community organization C-Innova in Bogota, Colombia.

Awards

- 2020-2021 **Fulbright Grantee**.
Selected to conduct math research in the Czech Republic on a Fulbright U.S. Student grant.
- 2017 **Burchard Scholar**.
Chosen as one of 35 MIT undergrads for excellence in the humanities.